

Lecture 12: Patents and Intellectual Property

HUM 4441: Engineering Ethics

Reaz Hassan Joarder, Samnun Azfar, and Yousuf Ibne Kamal Niloy

Junior Lecturer

{reazhassan, samnunazfar, yousufniloy}@iut-dhaka.edu

Department of CSE

Islamic University of Technology

“Black Armor with Light Sword”

The prompt was “black armor with light sword.” It names no film, no studio, and no character. What came back is a character owned by a major media company.

Generative AI can produce images or text very similar to copyrighted material, even when the prompt does not specifically request the material or its source.



GPT-4 response to the prompt “black armor with light sword.” Source:

G. Marcus and R. Southern, “Generative AI has a visual plagiarism problem,” *IEEE Spectrum*, 6 Jan. 2024.

“Animated Toys”



Another generic prompt returns another set of recognisable characters. This follows from how the systems are built. GPT-style models are trained by scraping online material, and what they absorb during training can resurface in what they generate.

Is producing output like this an unethical use of someone else’s intellectual property?

GPT-4 response to the prompt “animated toys.” Source: G. Marcus and R. Southern, “Generative AI has a visual plagiarism problem,”

IEEE Spectrum, 6 Jan. 2024.

Plagiarism?

Some authors and artists argue that this is **plagiarism** that threatens their **livelihood**.

- It allows content producers to “steal” ideas from genuinely creative people.
- There have been a number of **lawsuits**, including one by the *New York Times*.
- OpenAI has made agreements to **pay** some parties for the use of copyrighted material in a training set.

The **legal** answer is unsettled, and differs from country to country. So take the **ethical** question instead: does generative AI unethically use others' **intellectual property**?

Answering that means asking first what makes property worth respecting at all.

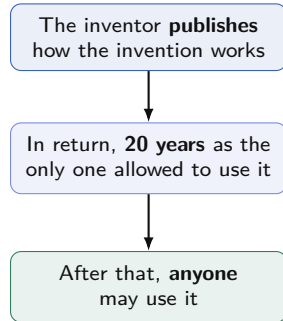
Is “Intellectual Property” Really Property?

The word **property** makes an idea sound like a car or a phone: something you own, and something someone else can take from you.

A patent does not work like that. It is a **trade**. The inventor publishes how the invention works, and in return gets **20 years** as the only one allowed to use it. After that, it is free for anyone.

And the **idea** by itself is never covered. A theorem, a plot, or a technique cannot be patented or copyrighted by anyone.

A patent is a temporary deal rather than permanent ownership.



The patent bargain: disclosure now, exclusivity for a while, then it belongs to everyone.

Why Taking Things Is Wrong: What if Everyone Did It?

The generalization test, from Lecture 11

An action is wrong if the reason for doing it stops working once everyone does it.

Stealing fails that test:

- The thief takes someone's property because he wants **possession and use** of it.
- But if no one respects property rights, the thief **won't have** possession and use either.
- In effect, there would be **no property** at all.

This works because physical goods are exclusive: if I have it, you do not.



A private property sign. Possession by one person rules out possession by another.

Copying Is Not Taking

But IP is **not** like other property: one can use IP **without denying others the use of it**.

Physical property

One car, one owner at a time.
Taking it removes possession
and use from someone else.

Intellectual property

Millions can possess and use the
same song, text, or design at once.
Nobody loses their copy.

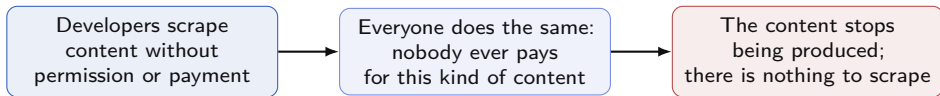
So the standard argument against theft does not, by itself, explain why copying is wrong.

Two students obtain the same \$80 textbook without paying. **A** takes a classmate's printed copy off her desk. **B** downloads a scanned PDF from a group chat, and the classmate keeps hers. Does the test condemn both, one, or neither? If B did something wrong, what argument are you using?

Scraping Fails a *Different Test*

Scraping is how these models get their training data: software automatically downloads text and images in bulk from public websites. Nobody is asked, and nobody is paid.

None of this says there is **no reason** to respect IP. A **second** generalization argument applies here, and it is the “garbage loop” pattern from last lecture: a practice that destroys what it feeds on.



On this argument it does not matter whether the output **resembles** the input, so the images we opened with are the wrong thing to be angry about.

What scraping takes is the money that pays for the reporting, which makes this a question about how journalism gets funded rather than about plagiarism.

The *New York Times* Lawsuit

The *Times* employs reporters who spend months on a single investigation. That work is expensive, is published online, and was used, without payment, in training sets. OpenAI has since agreed to pay some publishers for training material.

- On the argument just given, does the wrong depend on whether any generated text resembles a *Times* article?
- Would the same argument condemn scraping a public GitHub repository, a personal blog, or a student's course notes? What distinguishes the cases?

Intellectual Property Law

A brief introduction

Intellectual Property Law: The TRIPS Agreement

IP law differs in every country. The primary international treaty is the **TRIPS agreement**: Trade-Related Aspects of Intellectual Property Rights, dating from 1994 and administered by the **WTO**.

- It sets the **minimum** protection each member must provide. National law may go further, but not less far.
- The TRIPS minimums are life plus 50 years for copyright, and 20 years for a patent.
- Treaties set the standards; **national offices** grant and enforce the actual rights, such as the U.S. Patent and Trademark Office.



TRIPS is a floor, not a ceiling.

Four Kinds of Intellectual Property

Each protects something different, and they differ in how you obtain them and how long they last.

	What it protects	How you get it	How long it lasts
Copyright	A particular expression: code, text, image, melody	Automatically, on creation	Life + 50 years (TRIPS); 95 years for corporate works in the U.S.
Trademark	A mark identifying the source of goods or services	Registration, or commercial use	Indefinitely, while still used commercially
Patent	A new method, device, or material	Application and examination, in exchange for disclosure	20 years
Trade secret	Secret information giving commercial advantage	By keeping it secret	Until the secret escapes

Only copyright and patent expire on a clock. A trademark dies of disuse, a trade secret of disclosure.

Copyright

A **copyright** limits the number of copies others can make of a document or work of art without permission.

Duration in the U.S.

- For an individual: **life + 70 years**.
- For a corporate-owned work: **95 years** from publication.

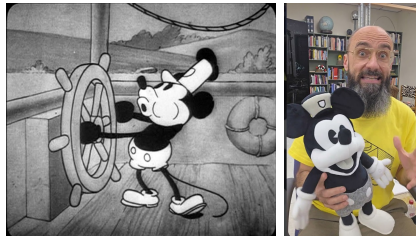


Copyright arises automatically on creation. Registration strengthens enforcement but does not create the right.

95 Years Later: Steamboat Willie

Steamboat Willie was published by Disney in 1928. As a corporate-owned work, its U.S. copyright ran for **95 years** from publication, and expired on 1 January 2024. Anyone may now copy, sell, or remake that film, and independent creators openly manufacture and sell 1928 Mickey merchandise without a licence from Disney.

What entered the public domain is only the **1928 version** of the character. Later designs stay under their own copyrights, and **trademark** still protects Mickey as a mark of Disney, for as long as Disney keeps using it commercially.



Left: Mickey in *Steamboat Willie* (1928), in the U.S. public domain since 1 January 2024 (Wikimedia Commons). Right: Michael Stevens with a *Steamboat Willie* plush that an independent creator made and sold without a licence from Disney (Vsauce, "I Just Committed NOT A Crime," YouTube, 23 July 2026).

What Copyright Does Not Cover

Ideas cannot be copyrighted. Copyright covers only a particular **expression** of ideas, such as a literary work or a musical composition.

Protected

The specific words, code, melody, or image you actually wrote or drew.

Not protected

The plot, the concept, the method, the underlying idea; anyone may write their own version.

This is why two people can write competing programs that do exactly the same thing without either infringing the other.

Fair Use

Is it legal to download this cartoon?

Copyright is not absolute. Certain uses of a copyrighted work are not infringement at all. Section 107 of the U.S. Copyright Act:

*"The **fair use** of a copyrighted work . . . for purposes such as criticism, comment, news reporting, **teaching** (including multiple copies for classroom use), scholarship, or research, is **not** an infringement of copyright."*

Whether a use is fair turns mainly on its **purpose**: commercial, or nonprofit and educational.

The same copy can be fair use in a classroom and infringement in a product.



"I KNOW IT'S ILLEGAL TO DOWNLOAD MUSIC FROM THE INTERNET... BUT YOU SAID THE STUFF I LISTEN TO ISN'T MUSIC!"

Cartoon © 2004 Randy Glasbergen, glasbergen.com. Reproduced here for classroom teaching.

Can Software Be Copyrighted?

Generally, **yes**.

- The TRIPS agreement explicitly says yes.
- But one **cannot** copyright the underlying idea or algorithm.

**Your source code is protected as an expression.
The technique it implements is not, so someone
else may reimplement it from scratch.**

```
/*  
 * Copyright (c) 2026 Example Ltd.  
 * All rights reserved.  
 *  
 * Permission is hereby granted, free  
 * of charge, to any person obtaining  
 * a copy of this software ...  
 */  
  
int sort(int *a, int n) {  
    /* the algorithm: not protected */  
}
```

The header and the code are a literary work and are protected.

The sorting method the function implements is not.

Trademark

A **trademark** is a word, phrase, symbol, or design that **distinguishes** the maker or source of goods. A **service mark** serves the same function for a service.

- Established by **registration**, or in common law by commercial use.
- Rights can last **forever**, as long as the mark stays in **commercial use**.
- One cannot claim a mark **already registered or in use** by another company in the same product class.



Marks identify the source of a product, so a buyer knows who is answerable for it.

Two Apples

Apple Corps, the Beatles' record label, registered its mark before Apple Computer existed. Both kept the name, because records and computers are different **product classes**. That held until Apple Computer started selling music, and the dispute ran from 1978 to 2007.

Patent

A patent is designed to encourage **disclosure** of ideas in exchange for a limited period of exclusive use.

- **20 years** in TRIPS.
- In the U.S.: **20 years**, except **15 years** for a design patent.

Can patent: a **new** method, device, or material.

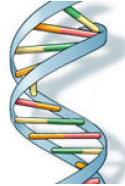


An invention already known to the public fails the novelty requirement and cannot be patented.

What Cannot Be Patented

- A pure idea, such as a theorem.
- Anything that occurs in nature.
- A “way of doing business,” even if automated by computer.
- “Look and feel,” for example of a spreadsheet.

Each exclusion keeps something available to everyone: mathematics, nature, ways of doing business, and interface conventions.



Naturally occurring sequences and the look and feel of an interface both fall outside patent protection.

Can Software or an Algorithm Be Patented?

Controversial, and it varies by country. The TRIPS agreement is **vague**. In the U.S. one can patent:

- a **software-based invention**;
- software that improves **computer functionality**;
- software that solves a problem arising only in the context of **computer technology**.

An algorithm **in principle** cannot be patented: it must be embodied in some device. **In practice** many have been, since a patent is issued not for source code but for a “series of steps,” which is an algorithm. **Bubble sort**, however, was denied a patent.

Ask a **patent lawyer**; it is a recognized specialty.

Patentable

“a faster way for a computer to do X”

Not patentable

“doing X, but on a computer”

Patentability turns on what problem the software is said to solve, not on the code itself.

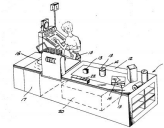
Who Owns a Patent?

A patent is **registered** in the name of the inventor.

The **owner** may be someone else, or a company. An employer normally **owns** any idea conceived by someone working **for hire**.

Being named as the inventor and owning the invention are two different things.

United States Patent		(10)	3,663,800
Myer et al.		(45)	May 16, 1972
(54)	OPTICAL LABEL READER AND DECODER		
(72)	Inventors: John H. Myer, Woodland Hills, Robert L. Hastings, Culver, Francis P. Williams, San Jose, Calif.		
(73)	Assignee: Hughes Aircraft Company, Culver City, Calif.		
(22)	Filed: Jan. 31, 1971		
(31)	Appl. No. 196,245		
(52)	Continuation of Ser. No. 718,534, Mar. 27, 1968, abandoned		
(51)	Int. Cl.		
(52)	Field of Search		
(53)	References Cited		
(54)	United States Patents		
(55)	Foreign Patents		
(56)	Continuation of Ser. No. 718,534, Mar. 27, 1968, abandoned		
(57)	ABSTRACT		
(58)	A coded label having a header, a unique identifiable word and data words coded thereon, a system for optically reading the label, including a reader for receiving light from the label, scanning a label from different incremental angles and generating prior signals in response to a code header, and a decoder coupled to receive the prior signals which is responsive to the header and to the unique identifiable word and the data words of the label, wherein the data will be decoded and displayed only after the label data words have been read, and a computer to read the data words.		
(59)	2,612,994 10/1952 Woodland		



US Patent 3,663,800, the supermarket barcode scanner. Field

[72] names the three inventors; field [73] gives the assignee,

Hughes Aircraft, which owns it.

Trade Secret

A **trade secret** is a secret formula or device that provides a commercial advantage. It can be bought, sold, and licensed.

It remains intellectual property **forever**, or until the secret gets out, as with the formula for Coca-Cola.

A patent gives up secrecy in exchange for 20 years of protection. A trade secret keeps the secrecy and gets no fixed term.



The Coca-Cola formula was never patented, since a patent would have published it.

Keeping a Secret Secret

There is one copy of the Coca-Cola formula, kept in a vault in Atlanta, USA.

- Two people have access to the vault.
- Their identities are secret.
- They are not allowed to travel together.
- When one is about to die, he or she chooses a successor.

The syrup is shipped as 9 components, combined at the factory.



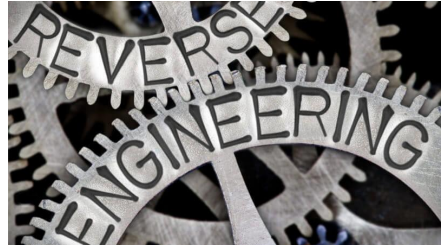
The Coca-Cola vault in Atlanta. The protection here is operational rather than legal, since the law cannot restore a secret once it escapes.

What Trade Secret Law Actually Prohibits

The law does not prohibit **use** of a trade secret. It only prohibits others from **stealing** one.

It is normally legal for another company to conceive the idea independently and use it. **Reverse engineering** is not theft; the idea was not really secret.

Trade secret law protects you against someone who steals the secret, not against a competitor who works it out independently.



Taking a product apart to learn how it works is lawful. Taking the documents that describe it is not.

Nondisclosure Agreements

A **nondisclosure agreement** prevents disclosure of proprietary information.

- The **duration** of the NDA is usually stated (1–5 years). If not stated, it may be forever.
- “Proprietary” is defined in the NDA itself.
- Disclosure of **trade secrets** at any time is **illegal**, with or without an NDA.
- Generally the company identifies what it considers to be trade secrets.



The NDA fixes the scope and the duration. Trade secret law applies regardless.

What Can You Take With You?

You finish a six-month internship where you signed a two-year NDA. At your next job you are asked to build something similar.

- You reuse a general design pattern you learned there. Permitted?
- You reuse a specific optimization the company identified as a trade secret. Permitted, and does the two-year limit help you here?
- You publish a blog post about a bug you found in their system, after the NDA expires. Where is the line between what you learned and what you took?

General skill and knowledge stay with you. Specific proprietary information does not.

Four Protections: Google Search

One product carries all four at once, on different parts of itself.

- **Copyright:** the source code of the search stack, and the artwork in each Doodle. Protected from the moment it is written, with no filing and no fee.
- **Trademark:** the “Google” wordmark and its four-colour lettering. It lasts as long as Google keeps using it commercially.
- **Patent:** PageRank, US 6,285,999. Filed 1998, granted 2001, expired 2019. It was granted to Stanford, which licensed it to Google.
- **Trade secret:** the ranking algorithm Search runs today. Never published, and with no expiry date.

Google published its 1998 ranking method in order to get the patent, and keeps the method it runs today secret, because it cannot do both.

Four Protections: A Phone

The same four sit on a device you are probably holding.

- **Copyright:** the operating system's source code, its icons, and its startup sound.
- **Trademark:** the maker's logo and brand name on the back, and the name of the phone model itself.
- **Patent:** the camera's image sensor and the antenna layout, both described in public applications.
- **Trade secret:** how the camera software turns the sensor's raw output into the picture you see. The sensor is patented and published; the tuning that makes photographs look the way they do is not.

You can check the first three yourself, on the product or in a patent register. The fourth you cannot, which is what keeping it secret buys.

Summary

- Generative AI can reproduce copyrighted material from generic prompts, because it is trained by scraping online content.
- The usual generalization argument against theft does not apply to IP: using it does not deny others its use.
- A specific generalization argument does apply: scraping without payment is wrong if the content would not exist had developers never paid for it, whether or not the output resembles the input.
- Copyright protects expression, not ideas; fair use exempts teaching, criticism, and research.
- Patents trade disclosure for 20 years of exclusive use; trademarks can last forever in commercial use; trade secrets last until the secret escapes.
- A patent is registered to the inventor, but work done *for hire* normally belongs to the employer.

References and Image Credits

J. Hooker, *AI and Intellectual Property*, Module 10 of *Ethical Issues in AI*, Carnegie Mellon University, CMU Osher, February 2024. Slide content in this lecture follows that module.

Images

- Generated images on the opening slides: G. Marcus and R. Southern, “Generative AI has a visual plagiarism problem,” *IEEE Spectrum*, 6 January 2024.
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